

Department of Electronics and Electrical Engineering

School of Engineering

Power Electronics

Lab-4

Single-Phase PWM Inverters

Student Name :

Matriculation Number :

Acknowledgements

This laboratory has been planned with the assistance and continuous support of Ian Young & Peter Miller, our power electronics technicians. Many Thanks! I would also like to acknowledge the work carried out by my predecessors, Vassilios Agelidis, Calum Cossar, Prof. Andrew Knox, Prof. Ravinder Dahiya and Dr Keliang Zhou originating and developing the Power Electronics 2 course.

Dr. Shuja Ansari

Glasgow, Feb 2021

The purpose of this lab is to simulate the performance of a Single-Phase Inverter with Voltage-Controlled Switches.

The following task has been adopted from Example 9.1 in the book by M. Rashid: "SPICE for Power Electronics and Electric Power". Please make sure to submit your script or design files for full marks.

Please work in a team of 2 people to complete this task. However, please make sure that you submit individual assignments.

A single-phase bridge inverter is shown in the figure below. The DC input voltage is 100 V. It is operated at an output frequency of f_o = 60 Hz with PWM control and four pulses per half cycle. The modulation index M = 0.6. The load is purely resistive with R = 2.5 Ω. Use PSpice to plot the:

(a) Instantaneous output voltage v_o.
(b) The instantaneous carrier voltage.
(c) The instantaneous reference voltage.

Use voltage-controlled switches to perform the switching action. Assume that A_r = 10 V, A_c = 10 × 0.6 = 6 V.

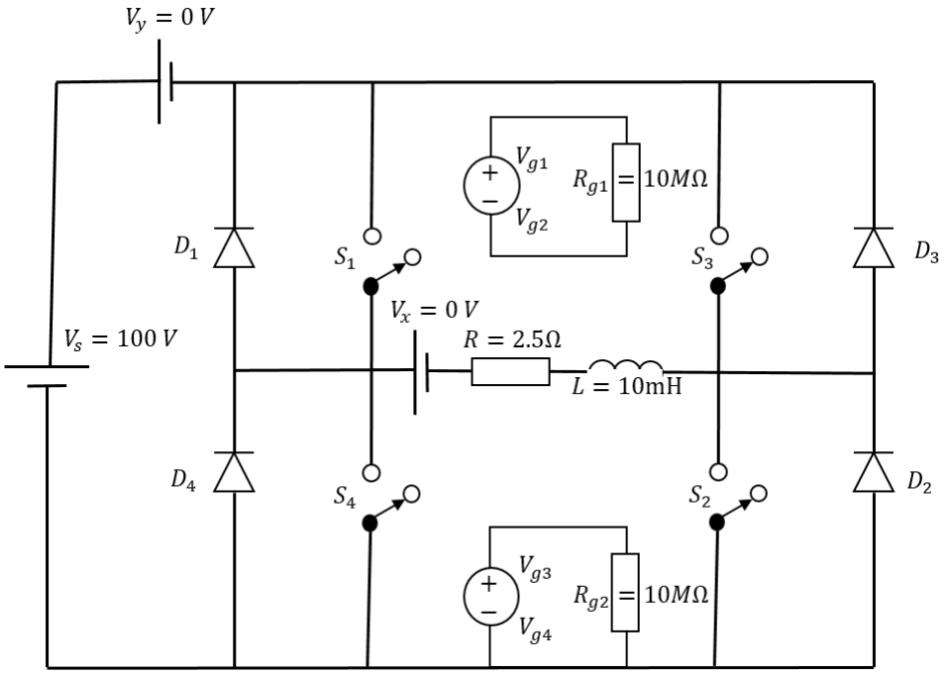


Figure 1: Bridge inverter for this lab exercise. Adapted from *M. Rashid: "SPICE for Power Electronics and Electric Power".*